# Prototypical Few-Shot Learning with HuBERT

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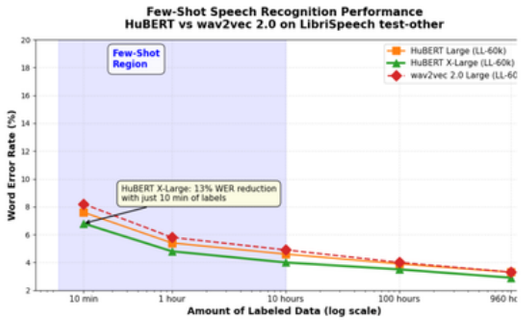
## Introduction

Audio understanding models typically require large labeled datasets (100h+). Even fine-tuning pretrained models often relies on tens to hundreds of hours of labeled data.

**Research question:**
**Can HuBERT achieve strong performance with only a few labeled examples?**

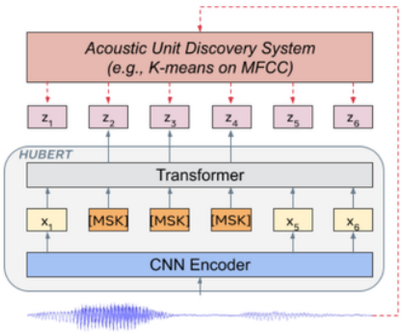
Goal:
Evaluate HuBERT's few-shot learning capability across diverse audio classification tasks in low-resource settings.




Illustrative results from prior work (LibriSpeech), shown for motivation.

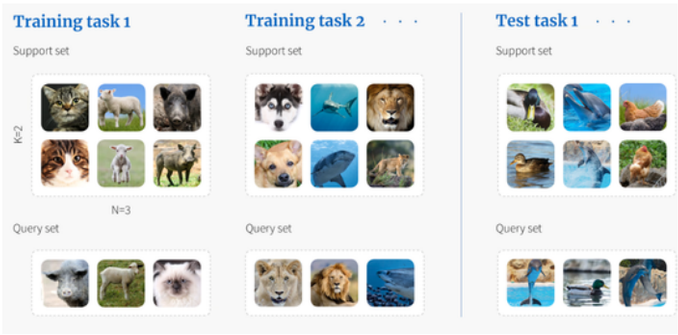
## HuBERT Model Overview

HuBERT is a self-supervised **foundation model** for speech representation learning. It learns from large-scale unlabeled speech by predicting discrete acoustic units.
The model combines a **CNN encoder** and a **Transformer**, enabling **strong transfer** to downstream tasks.




Overview of HuBERT pretraining

## Prototypical Method




## Results

### Main results

| Dataset | Inference 1s | Inference 5s | Inference 10s | Training 1s | Training 5s | Training 10s |
|---|---|---|---|---|---|---|
| Speech Commands | 55.7 | 81.1 | 86.4 | **87.0** | **93.1** | **93.7** |
| CREMA-D | 26.6 | 33.1 | 36.8 | **58.2** | **66.1** | **70.0** |
| TIMIT | 42.0 | 62.2 | 72.3 | **56.5** | **83.0** | **80.5** |
| Cats / Dogs / Birds | 72.8 | 92.1 | 95.4 | **100.0** | **100.0** | **100.0** |
| Snoring Detection | 55.0 | 62.8 | 66.8 | **100.0** | **100.0** | **100.0** |
| UrbanSound8K | 20.7 | 22.7 | 22.1 | **34.0** | **36.3** | **34.7** |
| Synthetic Noise (Low) | 99.4 | 99.7 | 99.9 | 98.9 | 98.9 | 98.9 |
| Synthetic Noise (Mid) | 96.9 | 98.3 | 98.4 | **95.3** | **97.5** | **97.2** |
| Synthetic Noise (High) | 91.1 | 95.5 | 96.2 | **93.6** | **99.4** | **99.0** |
| Synthetic Harmonics (Low) | 99.4 | 99.7 | 99.9 | 33.3 | 28.8 | 47.9 |
| Synthetic Harmonics (Mid) | 72.2 | 70.6 | 69.8 | 39.0 | 47.6 | 18.7 |
| Synthetic Harmonics (High) | 1.0 | 1.0 | 1.0 | **100.0** | **100.0** | **100.0** |

Few-shot accuracy for 1, 5 and 10-shot settings: inference averaged over 100 episodes, episodic prototypical training over 200 episodes. Synthetic datasets are reported at Low/Mid/High difficulty.

### Fine-tuning as a Reference (What Do We Gain ?)

| Dataset | Few-Shot Inference 1s | Few-Shot Inference 5s | Few-Shot Inference 10s | Episodic Prototypical Training 1s | Episodic Prototypical Training 5s | Episodic Prototypical Training 10s | Fine-Tuning FT |
|---|---|---|---|---|---|---|---|
| Cats / Dogs / Birds | 72.8 | 92.1 | 95.4 | **100.0** | **100.0** | **100.0** | 97.8 |
| Snoring Detection | 55.0 | 62.8 | 66.8 | **100.0** | **100.0** | **100.0** | **95.3** |
| UrbanSound8K | 20.7 | 22.7 | 22.1 | **34.0** | **36.3** | **34.7** | 14.6 |

Fine-tuning is performed using LoRA adapters to limit the number of **trainable parameters**

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Finally we have that :

- Episodic prototypical training performs best overall.
- Fine-tuning helps on simple datasets but fails on complex ones.
- Few-shot inference is already strong on small datasets.

## Conclusion and Takeways

- **HuBERT provides strong transferable representations** for **few-shot audio classification**, enabling competitive performance with **very limited labeled data**.
- **Episodic prototypical training** consistently **outperforms** both few-shot inference and fine-tuning across datasets.
- On **simple, well-separated datasets**, **few-shot inference** already reaches **near-optimal performance**, limiting the benefit of additional training.
- **Task complexity and acoustic diversity** largely explain performance differences across datasets.
- In terms of efficiency, few-shot inference runs in ~10 s, episodic training in 4–8 min, while fine-tuning can take ~1 h on UrbanSound8K, **making few-shot methods far more practical in low-resource settings**.

## References

- **Snell, J., Swersky, K., Zemel, R.** *Prototypical Networks for Few-shot Learning.*
- **Hsu, W.-N., Bolte, B., Tsai, Y.-H., Lakhotia, K.,Salakhutdinov, R., Mohamed, A.** *HuBERT: Self-Supervised Speech Representation Learning by Masked Prediction of Hidden Units.*
- **Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, L., Chen, W.** *LoRA: Low-Rank Adaptation of Large Language Models.*